

SIMPLE IMPACT STUDIES — SAMPLE DRAFT

Pinellas County Sample

Traffic Impact Study

Prepared September 24, 2026 · 27.9962, -82.7290



Simple Impact Studies — sample draft

Traffic Impact Study

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1.0 EXECUTIVE SUMMARY

This Traffic Impact Study has been prepared to evaluate the potential traffic impact, identify short-term roadway and circulation needs, determine potential mitigation strategies, and identify critical traffic issues that should be addressed during the planning process of the proposed Pinellas County Sample, located at Custom site (27.9962, -82.7290), Tampa MSA, within Pinellas County, Florida. The host controlling jurisdiction is Pinellas County (FDOT District 7 (Tampa)); the applicable review framework is Countywide Mobility Plan + Multimodal Impact Fee (transportation concurrency retired countywide). Analysis follows the FDOT Multimodal Transportation Site Impact Handbook (MTSIH, March 25, 2024) and the FDOT Quality/Level of Service Handbook v6.0 (August 2025); capacity analysis computes signalized control delay by the openly-published Webster (1958) uniform-delay formulation with an Akcelik overflow term — the basis of the capacity procedures FDOT Traffic Analysis Handbook §4.1 requires, which a submittal-grade study re-runs in licensed capacity software. The study covers 4 intersections within a 0.50-mile study area for land use 820 (Shopping Center / Retail) at a development size of 85 1,000 sqft GLA.

As part of this Traffic Impact Study, the following assignments were prepared consistent with the FDOT-approved methodology:

- Existing geometric conditions and assessment of the study impact area.
- Traffic data collection — daily volume counts and turning-movement counts at the critical study intersections and site driveways.
- Evaluation of existing traffic operations — Level of Service (LOS) and concurrency analysis.
- Traffic growth analysis, including committed-development assessment.
- Simulation of existing (current-year) and future conditions (opening year 2027) across three scenarios.
- Queue, turn-lane, and preliminary signal-warrant evaluation of the impacted network.

Findings:

- 2 intersections are projected to drop one or more LOS grades under Build conditions.
- 1 intersection operates at LOS E or F under Build conditions; mitigation per MTSIH 2024 §5 should be evaluated.

4	2	1	57.8s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

2.0 ANALYSIS METHODOLOGY

Per MTSIH 2024 §4.3, methodology and scope are established through a pre-application methodology meeting with the controlling FDOT District, county, and applicable MPO/TPO. Per MTSIH 2024 §4.3, the pre-application methodology meeting establishes scope. The methodology letter or meeting minutes must be included as Appendix A of the formal submittal.

2.1 Controlling Guidance

Primary references: FDOT Multimodal Transportation Site Impact Handbook (MTSIH), March 25, 2024; FDOT Multimodal Transportation Site Impact Applications Guide, June 5, 2024; FDOT Quality/Level of Service Handbook v6.0, August 2025; FDOT Policy 000-525-006 (Level of Service Targets for the State Highway System); FDOT Procedure 525-030-120 (project traffic forecasting); FDOT Traffic Analysis Handbook (TAH), October 2025.

2.2 Analysis Software

Per FDOT TAH §4.1, approved analysis tools are HCS, Synchro / SimTraffic, SIDRA INTERSECTION (roundabouts), CORSIM, and Vissim. This screening analysis applies the openly-published Webster/ Akcelik signalized-intersection model, reported in a format consistent with HCS output conventions. Vistro is not included in the FDOT TAH tool inventory; formal submittal output should be prepared in

HCS or Synchro.

2.3 Traffic Data Collection

Per MTSIH 2024, roadway-segment counts should be 72 consecutive hours (Monday afternoon through Friday morning) in urbanized, transitioning, and urban area classes, and 7 days in rural areas, in 15-minute increments on typical weekdays excluding holiday weeks. The default analysis peak per MTSIH 2024 §2.3.1 is the Weekday PM Peak Hour of Adjacent Street Traffic (one hour between 4–6 PM); MTSIH 2024 imposes no blanket Saturday-peak requirement for retail or restaurant land uses — midday, Saturday, or other special peaks are added only where site characteristics warrant (the Applications Guide fast-food case study analyzes AM + PM + midday). For turning-movement counts, MTSIH 2024 Appendix A (p. A-3) requires AM and PM TMCs covering trucks, pedestrians, and bicycles but does not prescribe duration or bin size; both are agreed at the pre-application methodology meeting. The Applications Guide Case Study 2 (§3.4.3) uses 8-hour TMCs (3 hr AM + 2 hr midday + 3 hr PM) as a worked example, while 2-hr AM and 2-hr PM in 15-minute bins is common Florida practice.

2.4 Analysis Scenarios and Time Horizons

Per MTSIH 2024 §4.3, minimum analysis years are: Existing, Future Background (No-Build), Future Build, and Future Build with Mitigation. This study is organized around three scenarios consistent with FDOT District practice: Scenario 1 — Existing Conditions (current-year); Scenario 2 — Future Conditions No-Build (opening year 2027); and Scenario 3 — Future Conditions Build (opening year 2027). Opening year is canonical; there is no fixed +5 horizon for concurrency or connection-permit work. Each year is explicitly labeled. For a Comprehensive Plan Amendment (CPA) review, the analysis must additionally include short-term (5-year) and long-term (10-year minimum) horizons. Pinellas County convention: Per local site-plan review under the countywide Mobility Plan; fee applies at permit (no horizon-year LOS test).

2.5 Trip Generation

Trip generation follows the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at the proposed development size of 85 1,000 sqft GLA. Net new external trips are calculated by applying pass-by and internal capture credits to gross trip generation per standard pass-by / internal-capture screening methodology. Per MTSIH 2024 §4.6.4, the fitted-curve equation is preferred when ≥20 data points are available, or when $R^2 \geq 0.75$ with the fitted curve falling within the data cluster and weighted standard deviation > 55% of the weighted average rate; otherwise the weighted average rate applies. Per MTSIH 2024 §4.6.6.6, pass-by trips at a site driveway cannot exceed 10% of the adjacent peak-hour two-way street traffic — this reasonableness check applies per roadway when the site fronts multiple streets and should be verified against the adjacent-street counts at submittal time.

Daily rate applied	80.00 trips per ksf × 85 ksf
AM peak rate applied	3.20 trips per ksf × 85 ksf
PM peak rate applied	8.00 trips per ksf × 85 ksf
Rate source	SANDAG 2002 “(Not So) Brief Guide of Vehicular Traffic Generation Rates for the San Diego Region” — Community Shopping Center (daily 80/ksf; PM pass-by 30%)
Pass-by capture applied	30%
Internal capture applied	0% (MTSIH 2024 §4.6.9 sets no statewide numeric cap; rate negotiated at the methodology meeting per NCHRP 684 / standard screening methodology)
Background growth applied	1.5% per year over 1 year(s)
Pinellas County TIA threshold	Per the controlling municipality's site-plan review procedures (Clearwater, St. Petersburg, Largo, and unincorporated Pinellas each administer review locally under the countywide Mobility Plan); multimodal impact fee assessed at permit
Weather condition	clear

Table: Site Trip Generation Summary

Period	Entering trips	Exiting trips
Daily	3,400	3,400
AM peak hour	166	106
PM peak hour	326	354

Table: Trip Generation by Period (raw → pass-by / internal capture → net external)

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	225	137	88
PM Peak Hour	680	204	0	426	204	222
Saturday Middyay	748	56	0	619	310	309
Daily Total	6,800	510	0	5,627	2,814	2,813

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross – pass-by – internal-capture) × auto-mode share (89.3% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

2.6 Growth Rate

Per FDOT TAH §2.7, demand projections should use the adopted regional MPO/TPO travel-demand model (TDM); where TDM use is not warranted, historical AADT trend growth from Florida Traffic Online (FTO) is the FDOT-wide convention. Background traffic is grown at 1.50% per year, applied as an explicit override, not derived from measured data. Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. The measured rate for this region is 2.92%/yr (FDOT TDA Annual_Average_Daily_Traffic_Historical FeatureServer layer 0 (per-year polyline snapshots; segment match by COSITE composite site ID) — median per-segment CAGR across 2129 matched count stations within the Tampa MSA bounding box (2021 → 2025)). The applied rate is not derived from measured data; its basis must be stated by the preparer. The override is not an FTO or FDOT TDA trend; for formal submittal, the FDOT District / FTO segment-level trend on the affected facilities is the authoritative input and should be confirmed at the methodology meeting.

2.7 Level of Service Standards

Per FDOT Policy 000-525-006, the peak-hour automobile-mode LOS standard on the State Highway System is LOS D in urbanized areas and LOS C in rural and transitioning areas. Constrained or backlogged facilities maintain their facility-specific designation. Roadway segment LOS reporting uses the FDOT Q/LOS Handbook v6.0 Generalized Service Volume Tables (GSVTs). Intersection LOS uses the conventional US signalized control-delay bands: A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s of average control delay per vehicle.

2.8 Context Classification

Per FDOT Q/LOS v6.0 (which replaced "complete streets" terminology with "context-based solutions") and FDM Chapter 200 §200.4 (Table 200.4.1), the study network's context classification (C1 Natural, C2 Rural, C2T Rural Town, C3R Suburban Residential, C3C Suburban Commercial, C4 Urban General, C5 Urban Center, C6 Urban Core) calibrates mode treatments and design standards; cross-section and lane widths follow FDM Table 210.2.1. The controlling context class should be confirmed against the FDOT Preliminary Context Classification mapping during the methodology meeting.

3.0 INTRODUCTION

The proposed Pinellas County Sample is located at Custom site (27.9962, -82.7290), Tampa MSA within Pinellas County, in the Tampa MSA area. The project consists of 85 1,000 sqft GLA under land use 820 (Shopping Center / Retail), and is anticipated to be completed by the opening year 2027.

Project name	Pinellas County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	27.9962°, -82.7290°

Region	Tampa MSA
Host jurisdiction	Pinellas County
FDOT District	FDOT District 7 (Tampa)
Review framework	Countywide Mobility Plan + Multimodal Impact Fee (transportation concurrency retired countywide)
Controlling document(s)	Pinellas County Multimodal Impact Fee Ordinance — Pinellas County Code Ch. 150 Art. II, fee schedule at Sec. 150-40 — administered countywide under the Forward Pinellas Mobility Plan. Ordinance update in progress: Phase 1 complete Aug 2025; Phase 2A (administrative) targeted Jun 2026; Phase 2B technical fee study anticipated fall/winter 2027.
Regional MPO / TPO	Forward Pinellas (unified MPO + Pinellas Planning Council — countywide land-use and transportation authority)
Opening year	2027

Surrounding land use, site plan figures, and detailed land-use description are dependent on the final site plan and are not produced by this screening tool. Final submittal should incorporate site plan figures (surrounding intersections and studied area) and a written project description per MTSIH 2024.

3.1 Study Area

The study area comprises the 4 intersections listed below, selected for their proximity to the site and their share of project-generated traffic. Existing lane geometry and traffic control at each location are shown in the Scenario 1 figures (see the turning-movement appendix).

#	Study intersection	Distance (mi)
1	Sunset Point Road	0.36
2	Sunset Point Road & US 19 Southbound Frontage Road	0.38
3	US 19 Northbound Frontage Road & Sunset Point Road	0.39
4	Sunset Point Road & Lawson Road	0.41

4.0 SCENARIO 1 — EXISTING CONDITIONS

Scenario 1 establishes existing (current-year) conditions from the collected daily and turning-movement counts at the study intersections. Existing Level of Service and control delay are summarized below; existing lane geometry and traffic control are shown in the Scenario 1 turning-movement figures (appendix).

Study intersection	Distance (mi)	Existing LOS	Existing delay (s)
Sunset Point Road	0.36	D	49.8
Sunset Point Road & US 19 Southbound Frontage Road	0.38	B	16.9
US 19 Northbound Frontage Road & Sunset Point Road	0.39	B	16.9
Sunset Point Road & Lawson Road	0.41	B	14.4

Existing AADT counts and functional classification are seeded from a live query of the FDOT Transportation Data and Analytics (TDA) ArcGIS REST services ([gis.fdot.gov RCI_Layers FeatureServer 0/3/15](https://gis.fdot.gov/RCI_Layers/FeatureServer/0/3/15) + `Access_Management_TDA`) within an 80-meter buffer of each intersection. Where the live extract returns a row, the in-table AADT, AADT year, and access-management class reflect the FDOT-published value at render time. Where no segment is matched (off-SHS local-road intersections, or radii beyond 80 m), values fall back to the engine estimate and must be confirmed against Florida

Traffic Online (<https://tdaappsprod.dot.state.fl.us/fto/>). Existing turn-lane storage length must be field-supplied as `existingStorageFt` on each intersection record to enable the §9.0 storage-bay-adequacy comparison; the renderer surfaces a deficit calculation only when that field is present.

4.1 FDOT TDA Site Snapshot (live)

RCI roadway ID	15150000
Begin/end mileposts	20.417 → 22.551
AADT	124,500 vpd (2025)
K factor (peak-hour)	9.000
D factor (directional)	54.300
Truck %	4.4%
Functional class	Urban Principal Arterial - Other
On State Highway System?	No
Access management class	Not classified — interim Rule 14-97.004(1) standards apply
FDOT District	D-7

Live extract from gis.fdot.gov/arcgis/rest/services/RCI_Layers/FeatureServer at render time. Direct on-segment match (≤60 m buffer). Cross-check against Florida Traffic Online (<https://tdaappsprod.dot.state.fl.us/fto/>) for the most recent annual update before submittal.

4.2 Roadway Segments Configuration

Physical and operational characteristics of each study segment from a live FDOT Transportation Data and Analytics (TDA) Roadway Characteristics Inventory (RCI) query at render time. Through-lane count and median type are RCI flat-file attributes not returned by this point query and must be confirmed from the RCI or a field inventory at submittal.

Study segment	RCI roadway	Functional class	SHS	Access cls	K	D
Sunset Point Road	15150000	Urban Principal Arterial - Other	No	—	9.000	54.300
Sunset Point Road & US 19 Southbound Frontage Road	15150000	Urban Principal Arterial - Other	No	—	9.000	54.300
US 19 Northbound Frontage Road & Sunset Point Road	15150000	Urban Principal Arterial - Other	No	—	9.000	54.300
Sunset Point Road & Lawson Road	15501000	Urban Minor Arterial	—	—	9.000	54.300

Live extract from gis.fdot.gov/arcgis/rest/services/RCI_Layers/FeatureServer. Access-management class per Rule 14-97 F.A.C.; functional classification per RCI Feature 121. Confirm lane count, median type, and posted speed from the RCI flat file or a field inventory before submittal.

5.0 SCENARIO 2 — FUTURE CONDITIONS NO BUILD (2027)

Scenario 2 represents future background (No-Build) conditions at the opening year 2027: existing volumes grown to the opening year without the proposed project. Background traffic is grown at 1.5% per year over 1 year, and any committed developments in the study area should be added to the No-Build network. The methodology and derivation of the applied growth rate are described in §2.6; the resulting No-Build Level of Service at each study intersection is reported in §7.0 alongside the Existing and Build scenarios.

Where the reviewing agency requires a travel-demand-model forecast (e.g., SERPM for Southeast Florida, or the controlling MPO/TPO model), the No-Build volumes should be reconciled against the adopted model run; the model version, base year, and horizon year are identified in the methodology letter. This screening analysis applies historical-AADT trend growth per FDOT TAH §2.7 in lieu of a model run.

Growth source: Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. The measured rate for this region is 2.92%/yr (FDOT TDA Annual_Average_Daily_Traffic_Historical FeatureServer layer 0 (per-year polyline snapshots; segment match by COSITE composite site ID) — median per-segment CAGR across 2129 matched count stations within the Tampa MSA bounding box (2021 → 2025)). The applied rate is not derived from measured data; its basis must be stated by the preparer..

5.1 AADT Growth Projection

Current-year segment AADT (live FDOT RCI, §4.2) is projected to the No-Build horizons at the applied 1.50%/yr compound growth rate derived from the FDOT historical-AADT trend (§2.6). These are screening-level trend projections; the adopted MPO/TPO travel-demand model volumes govern at formal submittal.

Study segment	Current AADT (yr)	CAGR	No-Build AADT (2027)	Design AADT (2047)
Sunset Point Road	124,500 (2025)	1.50%	128,263	172,752
Sunset Point Road & US 19 Southbound Frontage Road	124,500 (2025)	1.50%	128,263	172,752
US 19 Northbound Frontage Road & Sunset Point Road	124,500 (2025)	1.50%	128,263	172,752
Sunset Point Road & Lawson Road	28,000 (2025)	1.50%	28,846	38,852

Projection = current AADT × (1 + CAGR)^(horizon – count year). Where the live RCI count year is unavailable, the applied growth-years span is used. For submittal, use the per-segment FDOT historical AADT series (Florida Traffic Online) and, where required, the adopted MPO/TPO model volumes.

6.0 SCENARIO 3 — FUTURE CONDITIONS BUILD (2027)

Scenario 3 represents future Build conditions at the opening year 2027: the Scenario 2 No-Build network plus the proposed development's net new external trips at the assigned distribution. Per FDOT TAH §2.7, trip distribution and assignment should use the adopted regional MPO/TPO travel-demand model (Forward Pinellas (unified MPO + Pinellas Planning Council — countywide land-use and transportation authority)), with model version, base year, and horizon year identified in the methodology letter. This screening analysis distributes net new external trips to study-area zones with the Caltran gravity model (mass ÷ distance; see §6.1) and assigns them to signalized intersections within the study area; for formal submittal, distribution percentages and the TDM run identifier should be agreed upon during the methodology meeting.

6.1 Trip Distribution — Gravity Model

Project trips are distributed to the surrounding study-area zones with the gravity model used in the Caltran Engineering HCA Westside reference TIS, adopted here as the Florida distribution standard. Each zone attracts trips in proportion to its mass and inversely with its distance from the site — term = $M / (d^1 \cdot d_{\text{site}})$ — normalized so the zone shares sum to 100%. Zone mass M is the intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy; d_{site} (the site zone's own distance normalizer) is 1. The resulting shares set the directional distribution below and drive the project-trip assignment in §6.2.

Directional sector	Share of project trips				
Northeast (NNE + ENE)	0%				
Southeast (ESE + SSE)	0%				
Southwest (SSW + WSW)	100%				
Northwest (WNW + NNW)	0%				

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Sunset Point Road	SSW	0.36	2,500	7009.4	48.04%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Sunset Point Road & US 19 Southbound Frontage Road	SSW	0.38	990	2598.5	17.81%
US 19 Northbound Frontage Road & Sunset Point Road	SSW	0.39	990	2559.5	17.54%
Sunset Point Road & Lawson Road	SSW	0.41	990	2422.6	16.60%

Screening-grade gravity distribution over 4 study-area zones. For formal submittal the adopted regional MPO/TPO travel-demand-model distribution and the run identifier are confirmed at the methodology meeting per FDOT TAH §2.7; this gravity worksheet documents the screening basis.

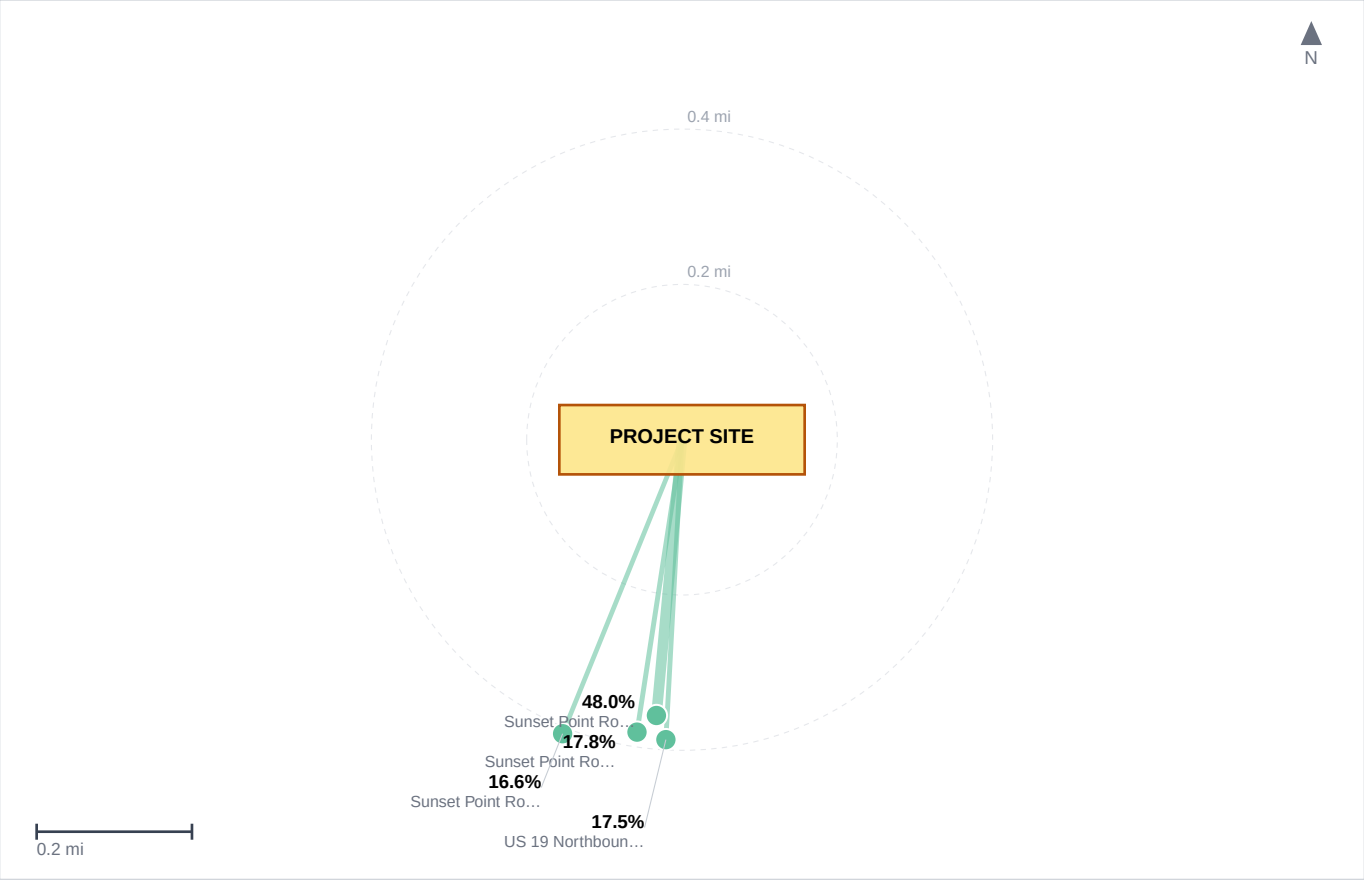
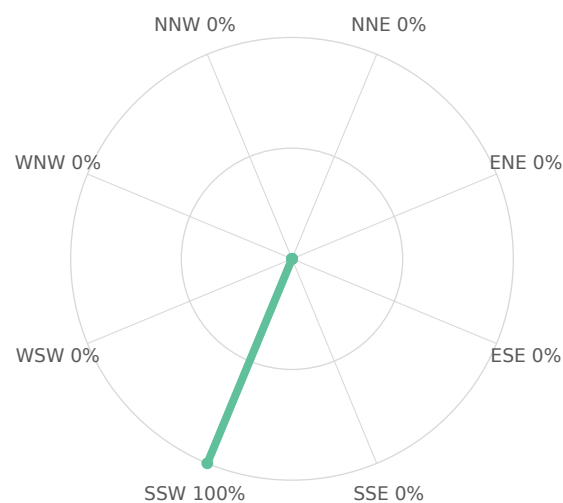


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

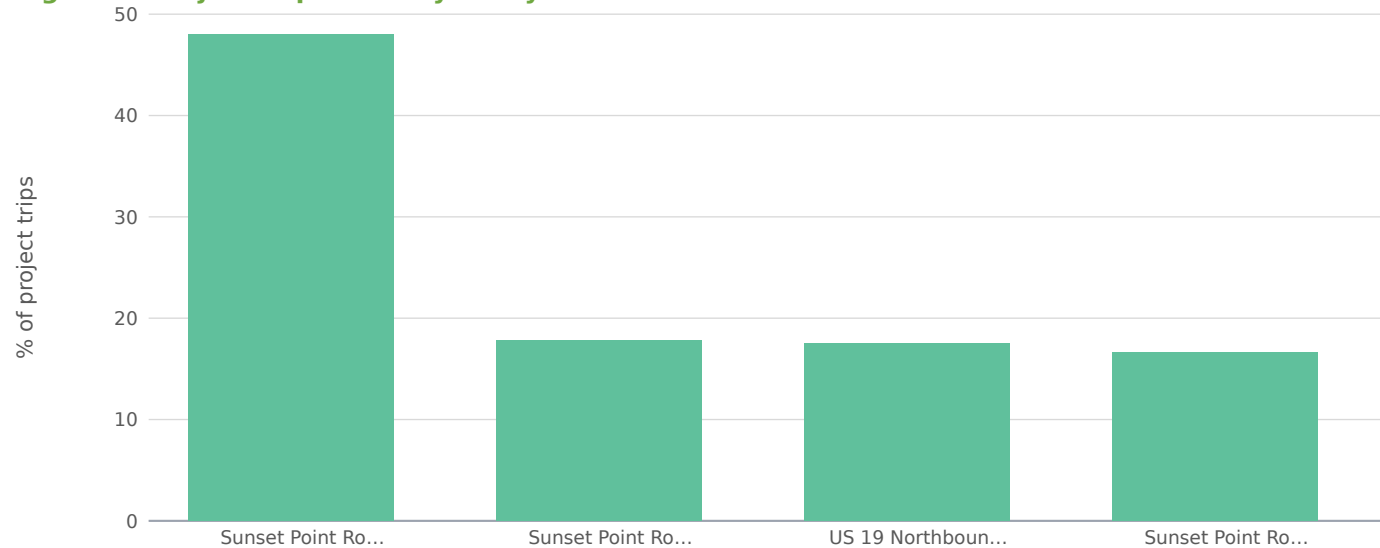
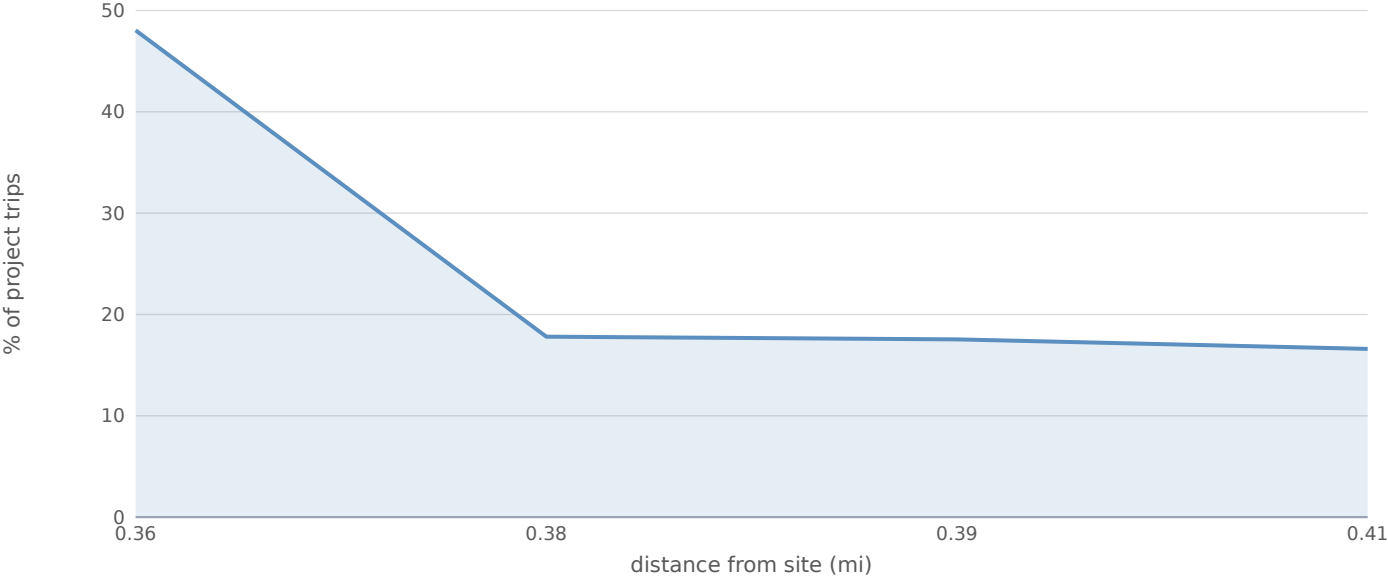


Figure — Trip Share vs. Distance from Site (Gravity Decay)



6.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Sunset Point Road	SSW	0.36	181	315	43.0%
Sunset Point Road & US 19 Southbound Frontage Road	SSW	0.38	69	102	13.9%
US 19 Northbound Frontage Road & Sunset Point Road	SSW	0.39	137	204	27.9%
Sunset Point Road & Lawson Road	SSW	0.41	44	111	15.2%

AM- and PM-peak project trips assigned to each study intersection from the §6.1 gravity distribution: the directional shares orient the loading toward the high-share sectors, while distance-decay from the site sets the magnitude that passes through each intersection. Entering/exiting splits at each intersection follow the site's period directional split.

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

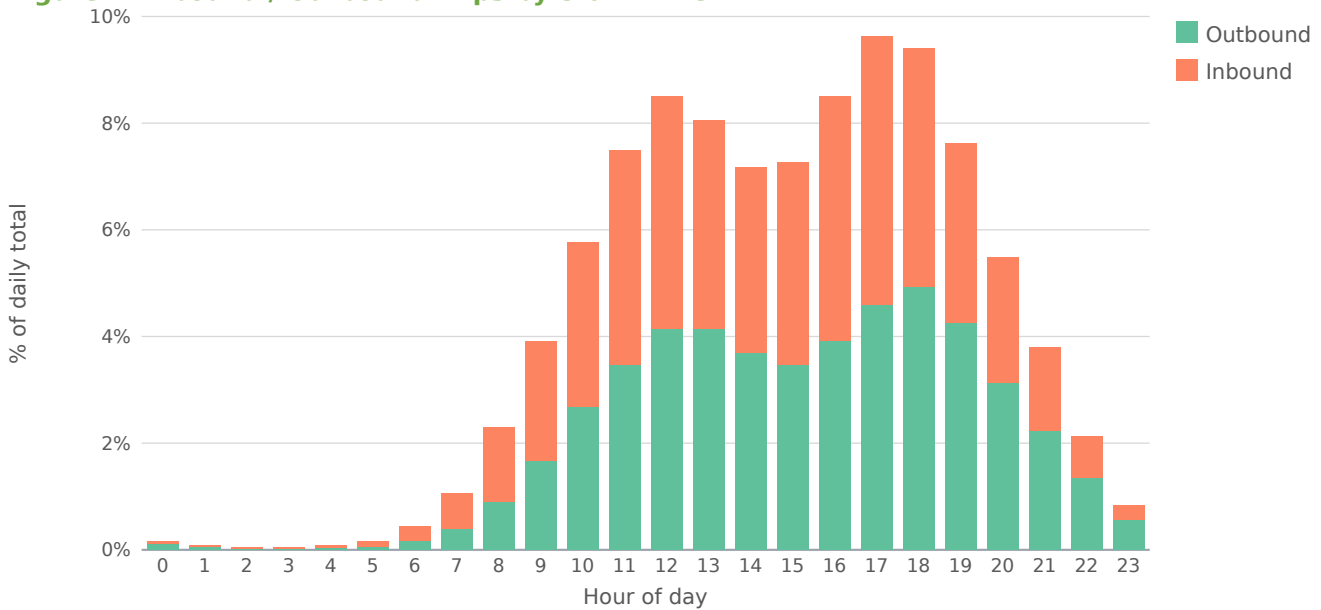
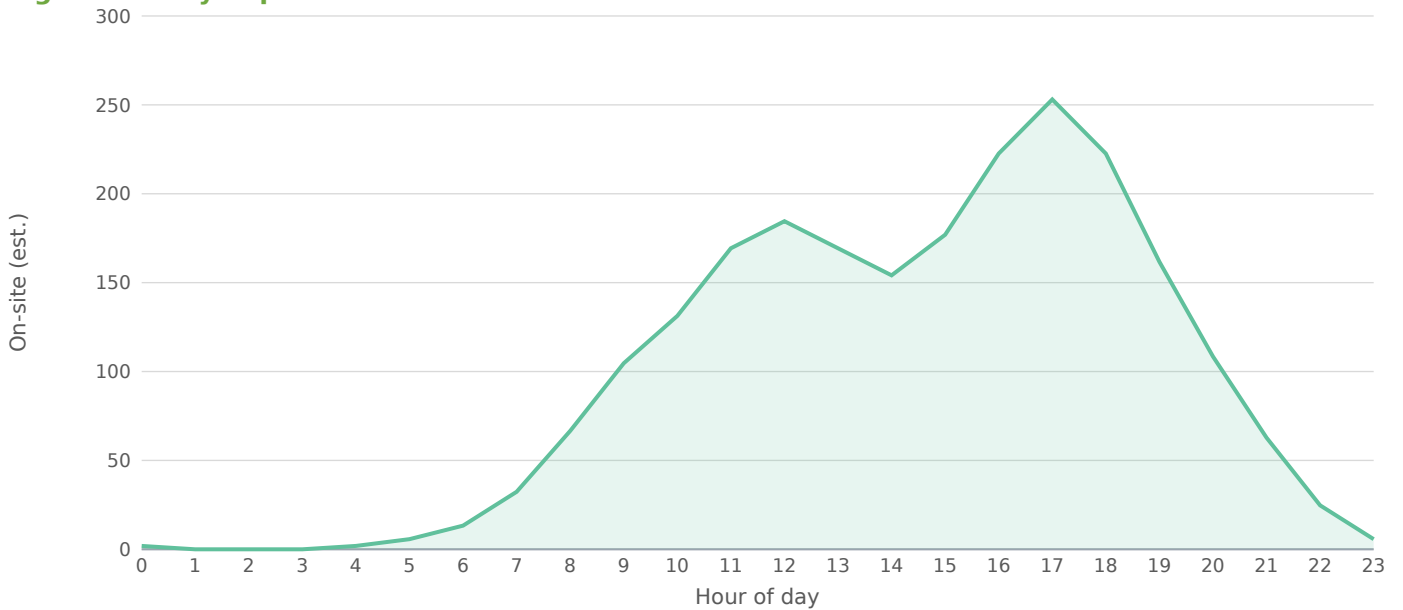


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

7.0 LEVEL OF SERVICE ANALYSIS

Intersection Level of Service is reported across the analysis scenarios against the conventional US signalized control-delay bands republished in state DOT traffic-engineering manuals. Four horizons are evaluated: (1) Existing — Scenario 1 current-year volumes; (2) No-Build — Scenario 2 at opening year 2027; (3) Build — Scenario 3 at opening year 2027; and (4) 20-Year Long-Range (2047) No-Build and Build. A ▲ flag marks any intersection projected to drop a LOS grade under Build conditions.

Intersection	Existing	Opening NB	Opening Bld	Design NB	Design Bld	Δ delay (s)
Sunset Point Road	D / 50	D / 53	▲ F / 111	F / 181	F / 258	57.8
Sunset Point Road & US 19 Southbound Frontage Road	B / 17	B / 17	B / 19	C / 25	C / 29	1.5
US 19 Northbound Frontage Road & Sunset Point Road	B / 17	B / 17	▲ C / 21	C / 25	D / 37	3.5
Sunset Point Road & Lawson Road	B / 14	B / 15	B / 16	C / 23	C / 27	1.3

Each scenario cell shows LOS grade / average control delay (s/veh) against the conventional US control-delay bands (A ≤10, B ≤20, C ≤35, D ≤55, E ≤80, F >80 s). Screening delays use each intersection's resolved signal timing — 3 from a Webster optimum cycle with Critical Movement Method splits derived from the no-build volumes; 1 at or beyond saturation on the generic 90 s / g/C 0.45 default (basis printed per intersection in the worksheets; timing held fixed across scenarios; approach capacity = 1,800 pc/h/ln × that phase's g/C) — and are superseded by a calibrated HCS/Synchro analysis of the actual lane geometry and signal timing at submittal.

7.1 Improvement Alternatives

The following intersection improvement alternatives are recommended to address projected Build-condition LOS impacts. Geometric mitigation should be designed to FDOT Design Manual (FDM 2026, Topic No. 625-000-002, dated January 1, 2026) standards — turn-lane warrants, deceleration / storage / taper lengths, intersection sight distance, and median-opening design per FDM Chapter 212; roundabouts per FDM Chapter 213. Proportionate-share, mobility-fee, or developer-contribution amounts for jurisdictions that retain concurrency (e.g., Miami-Dade Chapter 33-G) or operate mobility-fee programs (e.g., Hillsborough, Jacksonville/Duval Chapter 655, Miami-Dade Chapter 33E) should be calculated separately based on the controlling local-government ordinance.

Sunset Point Road [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +57.8s, design year +76.7s).

US 19 Northbound Frontage Road & Sunset Point Road [MINOR]

Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +3.4s, design year +12.1s).

7.2 Alternative Intersection LOS (with improvement)

The improvement alternatives in §7.1 are sized to mitigate the project's incremental impact and restore each affected intersection to approximately its No-Build (without-project) Level of Service. The table below states the with-improvement objective per location; the achieved Level of Service must be confirmed by a detailed HCS / Synchro analysis of the specific improvement at submittal.

Intersection	Build LOS	Δ delay (s)	Improvement class	Target LOS (improved)
Sunset Point Road	F	57.8	Major (geometry + signal)	D (verify — may remain constrained)
US 19 Northbound Frontage Road & Sunset Point Road	C	3.5	Minor (retiming)	B

Target LOS is the No-Build (without-project) grade the improvement is sized to restore, not a recomputed capacity result. Confirm the achieved delay / LOS with a detailed HCS / Synchro analysis of the specific geometry and signal timing at submittal.

8.0 QUEUE ANALYSIS (2027)

The 95th-percentile back-of-queue at each study intersection under Build conditions is summarized below. Queues are estimated from the standard cyclic-queue relation (Webster arrival/discharge form). The table below reports the governing approach; where imported Synchro records supplied measured turning-movement counts, §8.1 reports the queue for each left / through / right lane group

individually and compares it to that movement's own bay storage.

Intersection	Critical movement	Build Q95 (ft)
Sunset Point Road	Worst approach	1,507
Sunset Point Road & US 19 Southbound Frontage Road	Worst approach	368
US 19 Northbound Frontage Road & Sunset Point Road	Worst approach	368
Sunset Point Road & Lawson Road	Worst approach	327

8.1 Lane-Group Queues at Intersections with Measured Turning Movements

At the intersections below, an imported Synchro record supplied measured turning-movement counts, so the background traffic carries a real left / through / right split rather than an assumed one. Each lane group is analyzed on the same one-critical-lane screening basis as the approach (saturation flow $\times$ g/C), with the project's own trips assigned to specific movements by the path assignment; lane-group volumes cross-foot to their approach total. Where the imported record also carried a bay length, the queue is compared against that bay and a deficit is flagged. Where the record also carried LANE COUNTS, the Lanes column shows them and that group's capacity is sized as lanes $\times$ saturation flow $\times$ g/C; an em-dash in the Lanes column means no count was supplied and the group falls back to a single critical lane. Signal phasing is not carried by the import, so a calibrated Synchro / SimTraffic run supersedes these figures for design.

Intersection	Lane group	Lanes	Build vph	v/c	Q95 (ft)	Bay (ft)	Storage
Sunset Point Road	NB L	—	118	0.15	72	— —	
Sunset Point Road	NB T	—	1,355	1.67	1,386	— —	
Sunset Point Road	NB R	—	0	0.00	0	— —	
Sunset Point Road	SB L	—	0	0.00	0	— —	
Sunset Point Road	SB T	—	0	0.00	0	— —	
Sunset Point Road	SB R	—	0	0.00	0	— —	
Sunset Point Road	EB L	—	0	0.00	0	— —	
Sunset Point Road	EB T	—	0	0.00	0	— —	
Sunset Point Road	EB R	—	0	0.00	0	— —	
Sunset Point Road	WB L	—	0	0.00	0	— —	
Sunset Point Road	WB T	—	501	0.62	393	— —	
Sunset Point Road	WB R	—	118	0.15	72	— —	
Sunset Point Road & US 19 Southbound Frontage Road	NB L	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	NB T	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	NB R	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	SB L	—	76	0.10	32	— —	
Sunset Point Road & US 19 Southbound Frontage Road	SB T	—	427	0.57	225	— —	
Sunset Point Road & US 19 Southbound Frontage Road	SB R	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	EB L	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	EB T	—	534	0.71	303	— —	
Sunset Point Road & US 19 Southbound Frontage Road	EB R	—	76	0.10	32	— —	
Sunset Point Road & US 19 Southbound Frontage Road	WB L	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	WB T	—	0	0.00	0	— —	
Sunset Point Road & US 19 Southbound Frontage Road	WB R	—	0	0.00	0	— —	
Sunset Point Road & US 19 Northbound Frontage Road	NB L	—	0	0.00	0	— —	
Sunset Point Road & US 19 Northbound Frontage Road	NB T	—	529	0.71	301	— —	

US 19 Northbound Frontage	NB R	—	76	0.10	32	— —
Road & Sunset Point Road	SB L	—	0	0.00	0	— —
US 19 Northbound Frontage	SB T	—	0	0.00	0	— —
Road & Sunset Point Road	SB R	—	0	0.00	0	— —
US 19 Northbound Frontage	EB L	—	178	0.24	79	— —
Road & Sunset Point Road	EB T	—	432	0.57	227	— —
US 19 Northbound Frontage	EB R	—	0	0.00	0	— —
Road & Sunset Point Road	WB L	—	0	0.00	0	— —
US 19 Northbound Frontage	WB T	—	0	0.00	0	— —
Road & Sunset Point Road	WB R	—	0	0.00	0	— —
US 19 Northbound Frontage	NB L	—	64	0.10	30	— —
Road & Sunset Point Road	NB T	—	291	0.47	156	— —
US 19 Northbound Frontage	NB R	—	0	0.00	0	— —
Road & Sunset Point Road	SB L	—	0	0.00	0	— —
US 19 Northbound Frontage	SB T	—	291	0.47	156	— —
Road & Sunset Point Road	SB R	—	64	0.10	30	— —
US 19 Northbound Frontage	EB L	—	0	0.00	0	— —
Road & Sunset Point Road	EB T	—	0	0.00	0	— —
US 19 Northbound Frontage	EB R	—	0	0.00	0	— —
Road & Sunset Point Road	WB L	—	64	0.07	23	— —
US 19 Northbound Frontage	WB T	—	485	0.55	234	— —
Road & Sunset Point Road	WB R	—	64	0.07	23	— —

No lane group exceeds its imported bay storage under Build conditions.

9.0 TURN LANE EVALUATION (2027)

Turn-lane evaluation proceeds in two steps per FDM Chapter 212: first whether a turn lane is warranted at each affected approach (a function of the turning volume, opposing/through volume, and speed), and second — where a bay is warranted or already exists — whether its storage length is adequate for the projected queue.

9.1 Turn-Lane Warrants

A turn-lane warrant is a movement-level screen: an exclusive left-turn lane is evaluated against the FDM Chapter 212 / NCHRP 745 left-turn-lane guidelines (a function of advancing volume, opposing volume, and posted speed), and an exclusive right-turn lane is conventionally warranted where the peak-hour right-turn volume exceeds roughly 40–60 vph (or the applicable local threshold, e.g., Pinellas County land-development code). The proposed development adds approximately 326 inbound and 354 outbound trips in the PM peak hour, distributed to the site driveways and adjacent intersections. This screening decomposes those project-added trips into left / through / right movements at each study intersection — outbound trips enter on the site-facing leg and turn toward their destination sector, inbound is the mirror — and reports the result in the "Affected movements" table within each intersection's capacity worksheet, where the movement volumes cross-foot with that junction's added-trip total. At the 4 intersection(s) listed in §8.1 an imported Synchro record supplied measured existing turning movements, so the warrant screen there applies to the total movement volume (existing plus project). Everywhere else the screening does not carry a measured EXISTING turning-movement split: warrant thresholds apply to the total movement volume (existing plus project), and only the project increment is decomposed. A definitive turn-lane warrant determination — particularly at the site driveways connecting to the SHS — therefore remains subject to the AM/PM turning-movement counts and the trip distribution approved at the methodology meeting.

9.2 Storage-Bay Adequacy

Where a turn bay is warranted or already present, its storage length must contain the Build-scenario 95th-percentile queue without spill-back into the adjacent through lane; a deficit indicates the bay is shorter than the projected queue and requires either a bay extension (where adjacent infrastructure

permits) or a documented engineering finding that the spill-back is acceptable. Deceleration and taper lengths follow FDM Chapter 212.

Storage-bay adequacy: no intersection carries a field-measured existing storage length (``existingStorageFt``). Supply that field on each affected intersection record to enable the Required-vs-Existing-vs-Deficit turn-lane table per FDM Chapter 212 storage / taper standards. Turn-lane warrants at the site driveways must be evaluated against the assigned turning volumes at submittal.

10.0 CONCURRENCY ANALYSIS

Transportation concurrency was made optional statewide by HB 7207 (2011). For new development, the Development of Regional Impact (DRI) review track was further curtailed by SB 1216 / Ch. 2015-30 (which added F.S. 380.06(30) routing otherwise-DRI projects through the State Coordinated Review Process at F.S. 163.3184(4) in lieu of DRI), with cleanup completed by CS/CS/HB 1151 / Ch. 2018-158 — DRI is now a legacy branch retained only for amendments/abandonments of existing DRIs. Pinellas County review framework: Countywide Mobility Plan + Multimodal Impact Fee (transportation concurrency retired countywide). Controlling document(s): Pinellas County Multimodal Impact Fee Ordinance — Pinellas County Code Ch. 150 Art. II, fee schedule at Sec. 150-40 — administered countywide under the Forward Pinellas Mobility Plan. Ordinance update in progress: Phase 1 complete Aug 2025; Phase 2A (administrative) targeted Jun 2026; Phase 2B technical fee study anticipated fall/winter 2027.. LOS standard: Mobility-fee jurisdiction — no countywide vehicle LOS pass/fail; SHS LOS D urbanized per FDOT Policy 000-525-006 applies on State Highway System frontage for connection-permit review. Per Florida Statutes §163.3180(5) (h)1.a., local governments must consult with FDOT whenever a Strategic Intermodal System (SIS) facility is expected to be impacted by a comprehensive-plan amendment.

Note. Pinellas review is municipality-led: confirm the controlling city's own transportation-analysis submittal requirements (e.g., City of Clearwater or City of St. Petersburg development review) in addition to the countywide multimodal impact fee. Countywide plan amendments are analyzed by Forward Pinellas staff (MAX Index); applicant-prepared TIS work concentrates in municipal site cases.

10.1 Roadway Segment Capacity — Generalized Service Volumes

Roadway-segment level of service is screened against the FDOT Quality/Level of Service Handbook v6.0 (August 2025) Generalized Service Volume Tables (GSVTs). Per Q/LOS v6.0 the peak-hour two-way service volume is keyed to FDM Chapter 200 context class (C1-C6, C2T) and through-lane count, with D = 0.55 statewide. The applicable context class and lane count for each segment are FDOT Roadway Characteristics Inventory (RCI) attributes (Feature 126 context class; lane count from the RCI flat file) and must be confirmed during the methodology meeting. This screening tool selects the GSVT row only where those inputs are supplied and otherwise defers the segment v/c rather than assume a class.

Arterial (signalized) peak-hour two-way service volumes (vph) — Q/LOS v6.0 App. B:

Context / lanes	LOS C	LOS D	LOS E
C3C (Suburban Commercial), 2-lane undivided	1,380	1,950	—
C4 (Urban General), 2-lane	1,310	1,710	—
C3C, 4-lane divided	2,760	3,290	—
C4, 4-lane divided	2,070	2,980	—
C3R (Suburban Residential), 4-lane divided	3,090	3,360	—
C3C, 6-lane divided	4,290	4,870	—
C4, 6-lane divided	3,850	4,560	—
C2T (Rural Town), 2-lane undivided	1,310	1,710	—

Context / lanes	LOS C	LOS D	LOS E
C5 (Urban Center), 4-lane	2,350	3,450	3,870
C5, 6-lane	2,560	4,850	5,650
C6 (Urban Core), 4-lane	—	2,710	3,490
C6, 6-lane	—	4,960	5,350

C6 LOS C is undefined in v6.0 (C6 facilities are neither planned nor designed for auto LOS C); cells marked "—" past LOS D are F at signal capacity. Material adjustment multipliers (one-way $\times 1.2$; multilane without exclusive LT $\times 0.75$; 2-lane undivided without exclusive LT $\times 0.80$; non-State signalized $\times 0.90$; exclusive RT lane $\times 1.05$) and per-context K factors (C1/C2/C2T 8.5–10.5%; C3C/C3R/C4 7.5–9.5%; C5/C6 7.0–9.0%) apply per Q/LOS v6.0 Ch. 6. Freeway (Limited Access) GSVT LOS-D capacities: Urbanized 4/6/8-lane = 7,400 / 11,050 / 14,710 vph (AADT-equivalent 82,200 / 122,800 / 163,400); Rural 4-lane divided = 5,950 vph (AADT-equivalent 56,700).

Per-segment GSVT v/c is not computed for this run: a segment context class (C1–C6 / C2T) and through-lane count were not supplied. Provide them as inputs — or rely on the FDOT RCI data adapter once wired — to populate the roadway-segment LOS table against the service volumes above. Segment AADT available to the screen is listed in §4.0.

10.2 Site Access / Ingress-Egress

Per MTSIH 2024 §3.2 Table 5, driveway TIA-scoping is keyed to gross trips per day (including pass-by): Category A 1–20 vpd (single-family); B 21–600 (small multifamily / very small commercial); C 601–1,500 (small-mid retail / small office); D 1,501–4,000 (mid retail / mid office); E 4,001–15,000 (large retail / mixed-use); F 15,001–30,000 (very large mixed-use / mall); G $\geq 30,001$ (regional mall). Pre-application meeting + traffic study are required for Categories C–G (>600 vpd including pass-by). A connection-permit change-of-use is additionally triggered per F.S. 335.182(3)(b) when trip generation increases by >25% AND >100 vpd vs. the existing use. Connection to the FDOT State Highway System requires a connection permit per Rule 14-96 F.A.C. (last amended April 2, 2023). Driveway spacing, median-opening spacing, and signal spacing are governed by the access-management class (Classes 1–7) assigned to the impacted SHS segment per Rule 14-97 F.A.C. and FDOT Procedure 525-030-155; the class is stored in the RCI as Feature 146 / ACMANCLS (codes 00–07; 99 = unclassified, interim standards in Rule 14-97.004(1) apply until assignment). Driveway geometry (W, R, F, Y, G, Driveway Length, S, I; Categories A–D in FDM Chapter 214, Categories E–F–G punt to FDM Chapter 212), turn-lane warrants, deceleration-lane length, and intersection sight distance must be designed to FDOT Design Manual (FDM 2026) standards; off-SHS connections on city/county facilities follow the Florida Greenbook. The access-management class for the impacted SHS facility should be confirmed against the FDOT-published Access Management TDA layer.

11.0 TRANSIT AND MOBILITY

11.1 Transit Service

No transit stops detected within 0.25 mi of the site via the live Transit.land / OSM Overpass query. Verify against the controlling transit agency's published GTFS feed (e.g., Broward County Transit, Miami-Dade Transit, MARTA, LYNX, JTA, HART) at the methodology meeting. If transit-mode reduction is applied to trip generation, the supporting service must be cited and an alternative-mode trip reduction memo retained in the methodology letter (Appendix A or C per jurisdiction).

11.2 Internal Circulation and Multimodal

Internal site circulation, parking access, and service-vehicle pathways depend on the final site plan and are not included in this screening-level analysis. Internal queuing at the principal driveway should be evaluated for adequate storage between the SHS edge of pavement and the first internal conflict point per FDM guidance. Pedestrian and bicycle connectivity to the surrounding network and to the transit stops above should be confirmed against FDM Chapter 222/223 (sidewalk / bicycle facilities).

12.0 CRASH ANALYSIS

Site-radius crash query against the FDOT SSO Crashes (All) public FeatureServer returned no records within 0.5 mi over the available data window (FDOT public extract is comprehensive through

2018-2019; post-2019 records require Signal4 Analytics agency login). A formal submittal must include the 3-year Crash Analysis Reporting (CAR) extract for the affected intersections — this screening tool's public-data path cannot supply that for current years.

13.0 PRELIMINARY SIGNAL WARRANT ANALYSIS

A preliminary traffic-signal warrant screening is performed against the MUTCD (2009 Edition, FDOT-adopted per the Florida Traffic Engineering Manual) warrants — Warrant 1 (Eight-Hour Vehicular Volume), Warrant 2 (Four-Hour Vehicular Volume), Warrant 3 (Peak Hour), Warrant 4 (Pedestrian), Warrant 5 (School Crossing), Warrant 6 (Coordinated Signal System), Warrant 7 (Crash Experience), and Warrant 8 (Roadway Network). A definitive warrant determination requires the full 8-hour approach-volume counts and the applicable engineering study; a signal is not justified on any single warrant alone without an engineering study demonstrating that a signal will improve the overall safety and/or operation of the intersection.

Based on the projected Build-condition operations, the following study locations should carry a full signal / control-modification warrant analysis (Warrants 1–3 with 8-hour counts, plus Warrant 7 where the §12.0 crash review indicates a correctable pattern) at submittal:

Study location	Build LOS	Δ delay (s)	Preliminary indication
Sunset Point Road	F	57.8	Operational review warranted
US 19 Northbound Frontage Road & Sunset Point Road	C	3.5	Operational review warranted

Preliminary indication is a screening flag from projected LOS/delay only, not a warrant determination. The site driveways connecting to the SHS must additionally be evaluated for signalization / turn-lane control against the assigned turning volumes and the access-management class per Rule 14-97 F.A.C.

14.0 CONCLUSIONS AND RECOMMENDATIONS

The proposed Pinellas County Sample is projected to generate 6,800 daily trips (326 in / 354 out in the PM peak hour) at full build-out. Under Build (Scenario 3) conditions at opening year 2027, 2 study intersections are projected to drop a Level of Service grade and 1 operates at LOS E or F. The improvement alternatives identified in §7.1 should be advanced, and the applicable proportionate-share / mobility-fee obligation should be calculated per the controlling Pinellas County ordinance to maintain the FDOT SHS LOS standard within the study network.

14.1 Findings

- Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).
- Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).
- Auto-mode share of 89% applied to external trips for Tampa MSA; 11% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.
- Existing volumes were grown by 1.50%/yr over 1 year ($\times 1.01$) to the opening-year horizon.
- 4 signalized intersections fall within the study area; 2 are projected to drop at least one LOS grade in the opening year after build-out.
- 1 intersection is projected to operate at LOS E or F under the opening-year build condition and require formal mitigation per City of Tampa Transportation Division TIS guidance.
- Worst opening-year impact: Sunset Point Road — projected delay rises 57.8s (LOS D → F); 95th-pct queue 1507 ft on the critical approach.
- Largest projected delay change: 57.8s in the opening year (Sunset Point Road) and 76.8s in the design year (Sunset Point Road). The design-year figure is the larger; that difference is driven by background traffic growth over the design horizon, not by the development.
- Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the

formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a $\pm 0.5\%$ /yr background growth band around the applied value.

14.2 Programmed Projects

Committed-projects review should consult the FDOT Five-Year Work Program (<https://www.fdot.gov/workprogram>) and the controlling MPO/TPO Transportation Improvement Program (TIP) and Long Range Transportation Plan (LRTP). Programmed roadway and intersection improvements within the study area should be incorporated into the No-Build network. This screening analysis does not automatically integrate Work Program data; manual review is recommended for any submittal.

14.3 Professional Engineer Certification

A Florida TIS / MTIA deliverable must be signed and sealed by a Florida-licensed Professional Engineer per Florida Statutes Chapter 471 and Florida Administrative Code Rule 61G15-23.001. The cover and signature page of the formal submittal must bear the seal, signature, and date of the Engineer of Record.

14.4 Methodology Notes

- Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.
- Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.
- Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.
- Weather adjustment applies published rain/snow capacity-reduction factors as adopted in US traffic-engineering practice: clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.
- Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: the Caltran mass/distance gravity model — the Florida distribution standard (Caltran Engineering HCA Westside TIS) — allocates trips to surrounding zones by $T_j = (M_j / (d_j \cdot d_{site})) / \sum (M_x / (d_x \cdot d_{site}))$, where mass M_j is each signal's through-volume (destination-activity attraction proxy) and d_j is its straight-line distance from the site (site-zone distance normalizer $d_{site} = 1$). The normalized zone shares set the directional distribution and drive the project-trip assignment. Step 3 Mode Choice: a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.
- After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.
- Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.
- Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.
- Intersection-level control delay uses the standard signalized-intersection model $d = d_1 + d_2$ (Webster uniform delay + Akçelik time-dependent incremental-delay term, both openly published) with each intersection's own cycle length and green splits from the signal-timing resolver — a client Synchro record's measured cycle and per-phase splits where the record carries them, otherwise a Webster optimum cycle (60–120 s) with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006), a protected-left phase where the FHWA-HRT-04-091 cross product of left-turn and opposing through volume warrants one, and a pedestrian minimum green from the crossing width; timing is resolved once from no-build volumes and held

fixed across every scenario, approach capacity is $1,800 \text{ vphpl saturation flow} \times \text{that phase's } g/C (\times \text{weather factor})$, and an intersection at or beyond saturation ($Y \geq 0.85$) reports the flat 90s / g/C 0.45 screening default instead, 15-minute peak analysis period ($T = 0.25 \text{ hr}$) and pretimed-signal incremental-delay factor $k = 0.5$. Reported control delay is capped at 300 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

- Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (each leg from its own volume: the signal's design hour on the two main-road legs — its counted volume where the analyzer joined a compatible count, else the road-class baseline it assigned; each worksheet says which — half per direction, and half in the physical direction of a one-way carriageway; the road-class baseline on uncounted minor legs; a client link count where supplied — each worksheet states the source per leg. Turning movements are then balanced to the exit legs by iterative proportional fitting from a geometry seed, the refinement method of NCHRP Report 255 / NCHRP Report 765, so entries equal exits at every resolved junction; a junction that does not resolve to the road network keeps the screening allocation and says so) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (standard cyclic-queue relation with Poisson percentile factor: $Q_{95} \approx Q_1 \times 1.65 \times 25 \text{ ft/veh}$) are reported.

- Level of Service is assigned from the conventional US signalized control-delay bands republished in state DOT traffic-engineering manuals: A $\leq 10\text{s}$, B $\leq 20\text{s}$, C $\leq 35\text{s}$, D $\leq 55\text{s}$, E $\leq 80\text{s}$, F $> 80\text{s}$.

- Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a $\pm 0.5\%/yr$ growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

- Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

- Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as $(\text{turbo lanes} / \text{approach lanes}) \times (1 - g/C) / (g/C)$, with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%...+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

The Caltran mass/distance gravity model (the Florida distribution standard, §6.1) allocates trips to surrounding zones: $T_j = (M_j / (d_j \cdot d_{site})) / \sum (M_x / (d_x \cdot d_{site}))$. Zone mass M_j is each signal's through-volume (destination-activity proxy) and d_j its distance from the site ($d_{site} = 1$). The shares below are the §6.1 distribution.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Sunset Point Road	0.36	0.9	315	43.0%
US 19 Northbound Frontage Road & Sunset Point Road	0.39	0.9	204	27.9%
Sunset Point Road & Lawson Road	0.41	1.0	111	15.2%
Sunset Point Road & US 19 Southbound Frontage Road	0.38	0.9	102	13.9%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 89% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 11% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c: 0.64. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	1.49	426	0.64
Minor arterial	0.76	426	0.37
Principal arterial	0.33	426	0.26

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 2 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: $\sum \text{entering} = \sum \text{leaving}$ at 95 network junctions, max imbalance 0.00 veh. 4 study intersection(s) carry path-derived movements; 0 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c , control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Study scope: 4 signalized intersections lie within the 0.5-mile study area, and all 4 are carried as study intersections. The regional signal inventory holds 10 records inside that radius; 6 were merged into a junction already counted, because a signalized junction is commonly recorded once per approach rather than once per intersection. Merged records are duplicates of an analyzed intersection, not intersections omitted from the study.

Background approach volumes are resolved per leg — the signal's design hour on the main road (its counted volume where the analyzer had a compatible count, else the road-class baseline it assigned — each worksheet says which), the road-class baseline on uncounted legs, client link counts where supplied; half per direction, and half in the physical direction of a one-way carriageway — and the background turning movements in the diagrams are balanced to the exit legs by iterative proportional fitting (NCHRP 255/765 refinement) from a geometry seed; each worksheet states its leg sources and residual. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

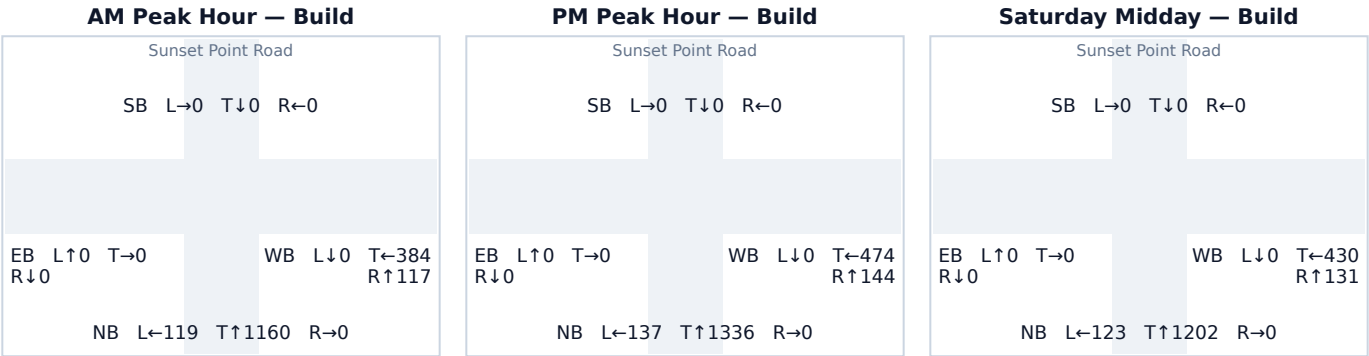
A.1 Sunset Point Road

Signal ID	tampa-2026005683
Location	SW Tampa · 0.36 mi from site
Added PM peak trips	315
Existing / No-Build (PM)	LOS D · 53.1 s/veh · v/c 0.99
Build (PM)	LOS F · 110.9 s/veh · v/c 1.16
Δ control delay	+57.8 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +57.8s, design year +76.7s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 0.99 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the analyzer's baseline volume for this signal (no compatible count — road-class or synthetic); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 8 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	1,134	117	0	1,250
SB approach	0	—	0	0	0
EB approach	0	0	—	0	0
WB approach	0	116	384	—	500
Σ exiting	0	1,250	500	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,269	204	1,473	1.57 → 1.82	285.4 → 300.0	F → F	1,507
SB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	508	111	618	0.63 → 0.76	22.6 → 27.5	C → C	534

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	204	65%
WB Through	111	35%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

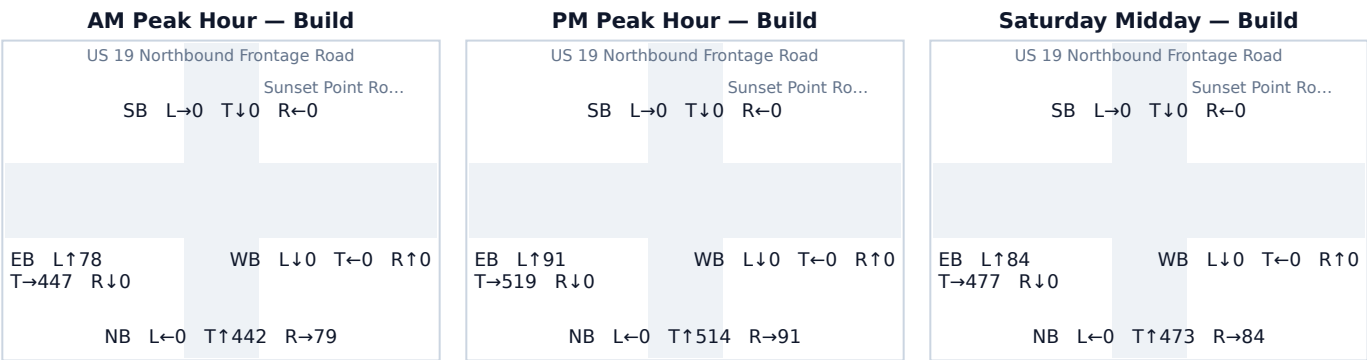
A.2 US 19 Northbound Frontage Road & Sunset Point Road

Signal ID	tampa-7288719554
Location	SW Tampa · 0.39 mi from site
Added PM peak trips	204
Existing / No-Build (PM)	LOS B · 17.1 s/veh · v/c 0.60
Build (PM)	LOS C · 20.6 s/veh · v/c 0.72
Δ control delay	+3.5 s/veh (LOS change)
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +3.4s, design year +12.1s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.41 / EW 0.42; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 2 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	421	0	75	495
SB approach	0	—	0	0	0
EB approach	0	75	—	425	500
WB approach	0	0	0	—	0
Σ exiting	0	495	0	500	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	502	102	605	0.67 → 0.81	19.1 → 24.8	B → C	366
SB	0	0	0	0.00 → 0.00	10.3 → 10.3	B → B	0
EB	508	102	610	0.67 → 0.81	18.9 → 24.5	B → C	368
WB	0	0	0	0.00 → 0.00	10.1 → 10.1	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	102	50%
NB Through	102	50%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

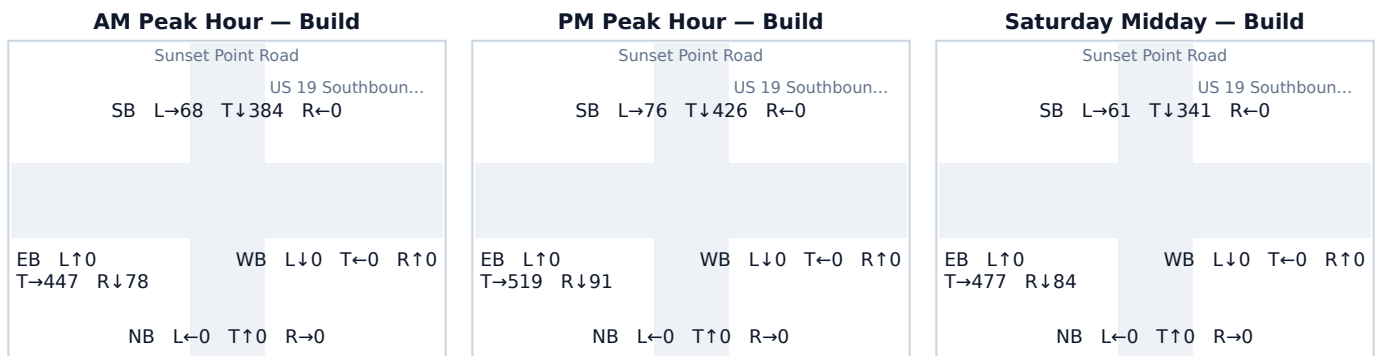
A.3 Sunset Point Road & US 19 Southbound Frontage Road

Signal ID	tampa-7288719556
Location	SW Tampa · 0.38 mi from site
Added PM peak trips	102
Existing / No-Build (PM)	LOS B · 17.1 s/veh · v/c 0.60
Build (PM)	LOS B · 18.6 s/veh · v/c 0.66
Δ control delay	+1.5 s/veh
Mitigation	No mitigation indicated at screening level (opening year +1.5s, design year +4.6s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.41 / EW 0.42; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 2 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	421	—	0	75	495
EB approach	75	0	—	425	500
WB approach	0	0	0	—	0
Σ exiting	495	0	0	500	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	10.3 → 10.3	B → B	0
SB	502	0	502	0.67 → 0.67	19.1 → 19.1	B → B	281
EB	508	102	610	0.67 → 0.81	18.9 → 24.5	B → C	368
WB	0	0	0	0.00 → 0.00	10.1 → 10.1	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	102	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

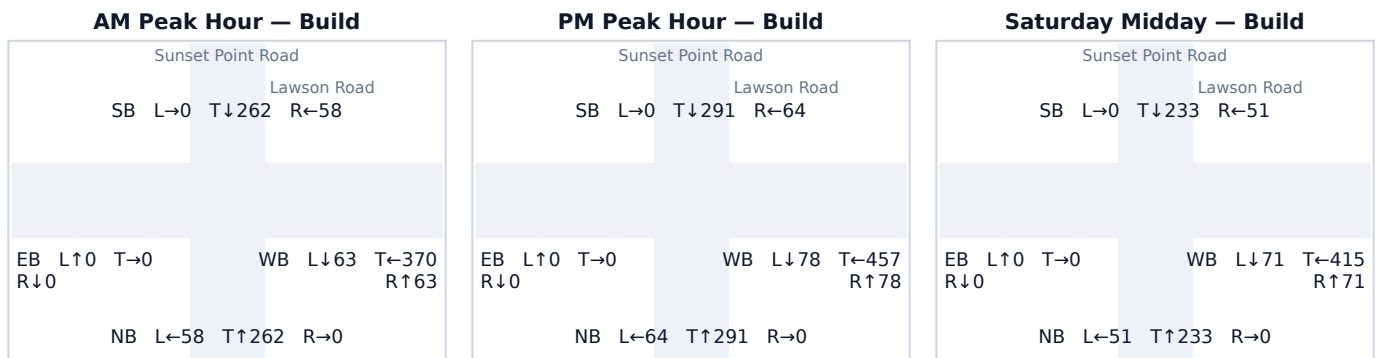
A.4 Sunset Point Road & Lawson Road

Signal ID	tampa-7520896874
Location	SW Tampa · 0.41 mi from site
Added PM peak trips	111
Existing / No-Build (PM)	LOS B · 14.6 s/veh · v/c 0.62
Build (PM)	LOS B · 15.9 s/veh · v/c 0.68
Δ control delay	+1.3 s/veh
Mitigation	No mitigation indicated at screening level (opening year +1.4s, design year +4.8s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.34 / EW 0.49; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 5 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	287	63	0	350
SB approach	287	—	63	0	350
EB approach	0	0	—	0	0
WB approach	63	63	369	—	495
Σ exiting	350	350	495	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.57 → 0.57	19.8 → 19.8	B → B	199
SB	355	0	355	0.57 → 0.57	19.8 → 19.8	B → B	199
EB	0	0	0	0.00 → 0.00	7.9 → 7.9	A → A	0
WB	502	111	613	0.57 → 0.70	13.6 → 16.5	B → B	327

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	111	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.